

LA BAIE, QC, Canada

WMO: 713880

Lat: 48.2981N

Lon: 70.9194W

Elev: 152

StdP: 99.52

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.3	-27.6	-34.6	0.1	-29.9	-31.9	0.2	-27.5	12.0	-14.4	11.1	-15.1	2.3	150	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	29.0	19.6	27.0	19.1	25.2	18.3	21.6	26.3	20.5	24.8	19.5	23.5	4.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.1	23.7	19.1	14.1	22.5	18.0	13.2	21.5	63.8	26.0	60.0	25.1	56.5	23.4	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.0	8.0	-34.7	32.9	3.2	1.7	-37.0	34.1	-38.9	35.1	-40.7	36.1	-43.1	37.3		
(5)			-34.8	23.9	3.2	1.3	-37.1	24.8	-39.0	25.5	-40.8	26.3	-43.1	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-15.0	-13.1	-6.9	1.8	10.3	15.1	17.9	17.1	12.6	5.6	-1.3	-9.8		
	DBStd	12.64	7.77	7.10	6.70	4.53	4.37	3.78	2.91	3.02	4.17	4.11	5.12	6.94		
	HDD10.0	3365	777	646	523	246	51	4	0	0	20	147	339	613		
	HDD18.3	5688	1035	879	781	496	252	109	43	60	178	394	588	871		
	CDD10.0	794	0	0	0	1	59	158	246	221	97	12	1	0		
	CDD18.3	75	0	0	0	0	2	14	31	22	6	1	0	0		
	CDH23.3	848	0	0	0	1	83	191	297	209	66	3	0	0		
	CDH26.7	203	0	0	0	0	22	51	73	46	12	0	0	0		
	WSAvg	3.4	3.6	3.7	3.9	3.9	3.5	3.1	2.8	2.7	2.9	3.3	3.5	3.4		
(15) Precipitation	PrecAvg	958	62	53	57	59	80	91	116	101	102	85	77	75		
	PrecMax	1220	122	112	128	106	181	160	320	245	226	149	147	142		
	PrecMin	754	17	13	12	12	15	19	37	35	41	27	22	27		
	PrecStd	105	24	23	26	25	33	32	45	41	36	29	26	28		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	4.1	9.7	18.4	29.4	31.1	31.1	30.7	28.3	21.8	14.1	5.0		
		MCWB	2.8	2.0	5.3	10.4	16.1	20.3	21.2	21.1	20.0	15.3	10.9	3.3		
	2%	DB	1.7	1.4	5.9	14.4	25.6	27.9	28.7	27.7	25.2	17.9	9.4	2.4		
		MCWB	1.1	0.1	2.5	7.4	14.9	18.4	20.3	20.0	19.0	13.5	7.2	1.3		
	5%	DB	-0.6	0.1	4.0	11.2	22.2	25.4	26.5	25.6	22.4	14.2	7.0	1.0		
		MCWB	-1.4	-0.9	1.5	6.1	13.5	17.7	19.7	18.9	17.2	10.9	5.6	0.2		
	10%	DB	-3.5	-1.7	2.4	8.5	19.1	23.1	24.6	23.7	20.0	11.9	5.2	-0.2		
		MCWB	-4.4	-2.7	0.7	4.6	12.2	16.2	18.7	18.1	16.1	9.5	3.9	-0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.7	6.9	11.7	19.2	22.4	23.4	22.5	21.7	16.5	11.2	3.6		
		MCDB	3.6	3.4	8.6	16.8	25.3	28.5	28.4	27.7	26.5	20.8	13.8	4.5		
	2%	WB	1.1	0.4	3.2	8.2	16.7	20.4	21.7	21.1	19.8	13.6	7.7	1.5		
		MCDB	1.6	1.3	5.0	12.8	22.6	25.2	25.9	25.8	23.5	17.1	9.0	2.2		
	5%	WB	-1.4	-0.9	2.0	6.6	14.4	18.9	20.7	20.0	18.0	11.4	5.6	0.4		
		MCDB	-0.7	0.1	3.6	10.3	20.3	23.3	24.9	23.9	21.7	13.6	6.7	1.0		
	10%	WB	-4.3	-2.8	0.8	5.1	12.8	17.6	19.6	19.0	16.5	9.7	3.9	-0.9		
		MCDB	-3.5	-1.7	2.2	8.0	18.2	21.6	23.4	22.4	19.3	11.6	5.1	-0.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.2	11.1	9.7	12.5	12.0	11.5	11.6	11.3	8.4	6.8	8.2		
		MCDBR	10.8	9.7	12.3	14.8	19.1	16.6	15.1	15.1	15.4	13.2	9.3	7.9		
		MCWBR	10.3	8.8	9.6	9.5	9.6	8.8	8.0	8.2	9.3	9.0	7.7	7.5		
	5% WB	MCDBR	10.6	9.3	9.8	12.8	15.6	13.2	12.9	12.4	13.5	12.0	8.6	7.8		
		MCWBR	10.2	8.6	8.0	9.0	8.8	8.0	7.7	7.3	8.7	8.7	7.4	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.284	0.316	0.382	0.363	0.385	0.405	0.385	0.372	0.330	0.313	0.267		
	taud		2.273	2.225	2.198	2.189	2.396	2.365	2.320	2.380	2.403	2.505	2.395	2.322		
	Ebn at Noon		807	870	895	864	899	877	852	855	831	818	742	756		
	Edh at Noon		73	97	119	134	113	118	122	110	98	75	67	62		
(44) All-Sky Solar Radiation	RadAvg		1.31	2.27	3.51	4.45	5.14	5.49	5.20	4.43	3.40	1.95	1.07	0.87		
	RadStd		0.16	0.30	0.29	0.35	0.50	0.37	0.51	0.29	0.26	0.20	0.11	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page